B. Cosmic Tables

time limit per test 3 seconds
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1300 | Tags: data structures, implementation | https://codeforces.com/problemset/problem/222/B

The Free Meteor Association (FMA) has got a problem: as meteors are moving, the Universal Cosmic Descriptive Humorous Program (UCDHP) needs to add a special module that would analyze this movement.

UCDHP stores some secret information about meteors as an $n \times m$ table with integers in its cells. The order of meteors in the Universe is changing. That's why the main UCDHP module receives the following queries:

- The query to swap two table rows;
- The query to swap two table columns;
- The query to obtain a secret number in a particular table cell.

As the main UCDHP module is critical, writing the functional of working with the table has been commissioned to you.

Input

The first line contains three space-separated integers n, m and k ($1 \le n, m \le 1000$, $1 \le k \le 500000$) — the number of table columns and rows and the number of queries, correspondingly.

Next n lines contain m space-separated numbers each — the initial state of the table. Each number p in the table is an integer and satisfies the inequality $0 \le p \le 10^6$.

Next k lines contain queries in the format "$s_i\ x_i\ y_i$", where s_i is one of the characters "c", "r" or "g", and x_i, y_i are two integers.

- If s_i = "c", then the current query is the query to swap columns with indexes x_i and y_i ($1 \le x, y \le m$, $x \ne y$);
- If s_i = "r", then the current query is the query to swap rows with indexes x_i and y_i ($1 \le x, y \le n$, $x \ne y$);
- If s_i = "g", then the current query is the query to obtain the number that located in the x_i-th row and in the y_i-th column ($1 \le x \le n$, $1 \le y \le m$).

The table rows are considered to be indexed from top to bottom from 1 to n, and the table columns — from left to right from 1 to m.

Output

For each query to obtain a number (s_i = "g") print the required number. Print the answers to the queries in the order of the queries in the input.

Examples

Input

```
3 3 5
1 2 3
4 5 6
7 8 9
g 3 2
r 3 2
c 2 3
g 2 2
g 3 2
```

Output

```
8
9
6
```

Input

```
2 3 3
1 2 4
3 1 5
c 2 1
r 1 2
g 1 3
```

Output

```
5
```

Note

Let's see how the table changes in the second test case.

After the first operation is fulfilled, the table looks like that:

2 1 4

1 3 5

After the second operation is fulfilled, the table looks like that:

1 3 5

2 1 4

So the answer to the third query (the number located in the first row and in the third column) will be 5.